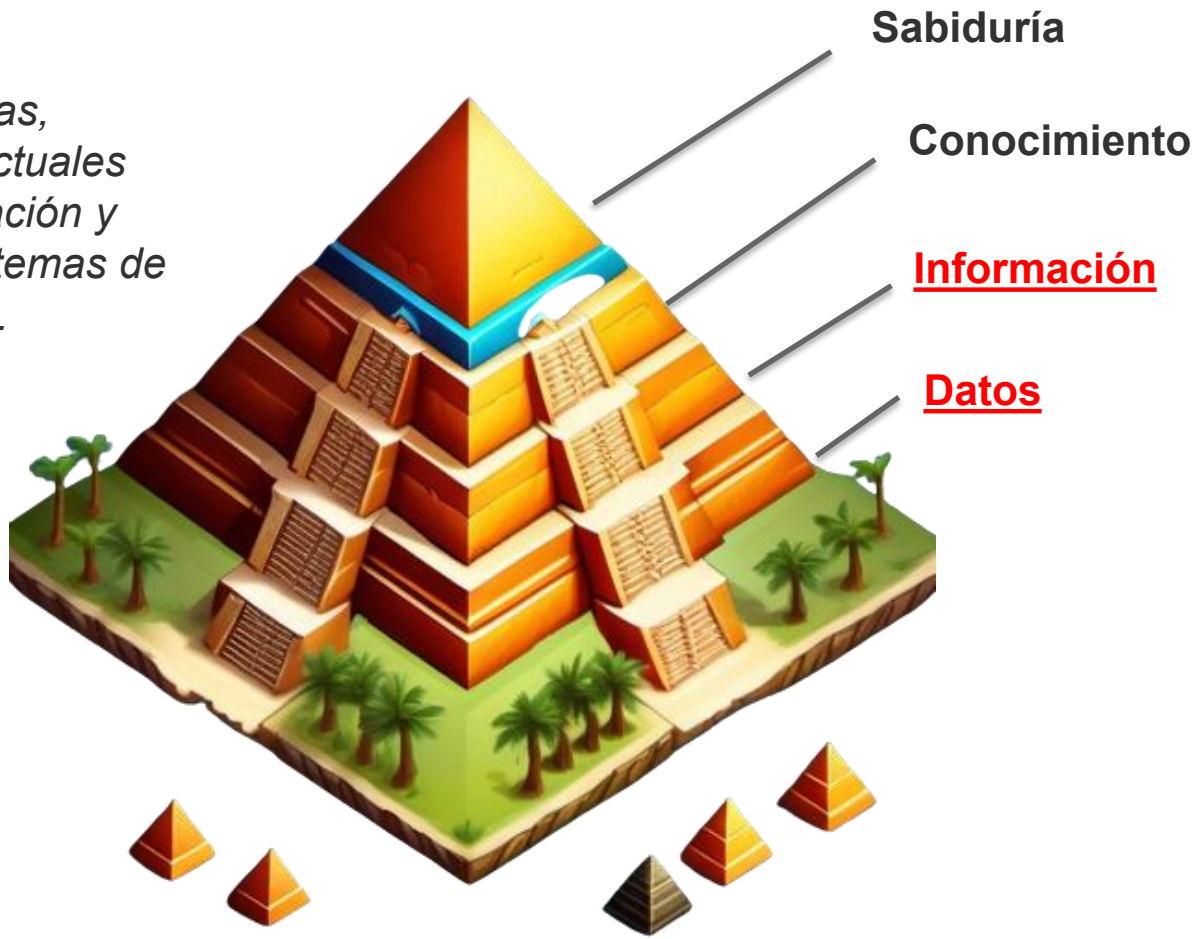


GESTIÓN DE SISTEMAS DE DATOS MASIVOS

Objetivos

Comprender las teorías, modelos y técnicas actuales para la gestión, utilización y administración de sistemas de almacenamiento de…



Information Management

01. Need for Information Transformation

02. ETL-Based Mechanisms

03. ELT-Based Mechanisms

04. Classic Transformations

05. Advanced/Deep Transformations

Need for Information Transformation

- **Main propose:**
  - **Integration with a Tool/Library Outside the System** : In some scenarios, we need our data to be in a specific format for use in external tools such as web service calls. In other cases, it's the processing libraries that impose the input format that our data must conform to..
  - **As part of the processing process :** In certain cases, data processing is not as straightforward and requires a complex transformation of the data.
  - **Export/Load into an External System**: When the final result of data loading and the transition to information ends, from our perspective, with a transfer to an external system, changes are required to ensure concepts such as security, privacy, and adaptation (similar to the concept of integration with an external tool) before offering it to a system outside of our own.

Need for Information Transformation

- **Integration with a Tool/Library Outside the System (example Rest-Service)**

If we want to work with a mapping system using Google Maps, we will require the field sent to the Google API system to be a two-dimensional array with latitude and longitude, with maximum precision, and the product key inserted for correct functionality. If this process is occasional, it can be done directly during execution. However, if it involves multiple points on a map, it's always better to have these repetitive calculations stored as part of the system.

Need for Information Transformation

- **As part of the processing process (example NER)**

The NER (Named Entity Recognition) process is the transformation of our text-format data into a complex conversion to detect entities and capture parts of them as part of our problem's requirements. This process transforms our data (or extends it) into a new set of data, potentially more enriched for our ultimate goal.

Need for Information Transformation

- **Export/Load into an External System (example database warehouse)**

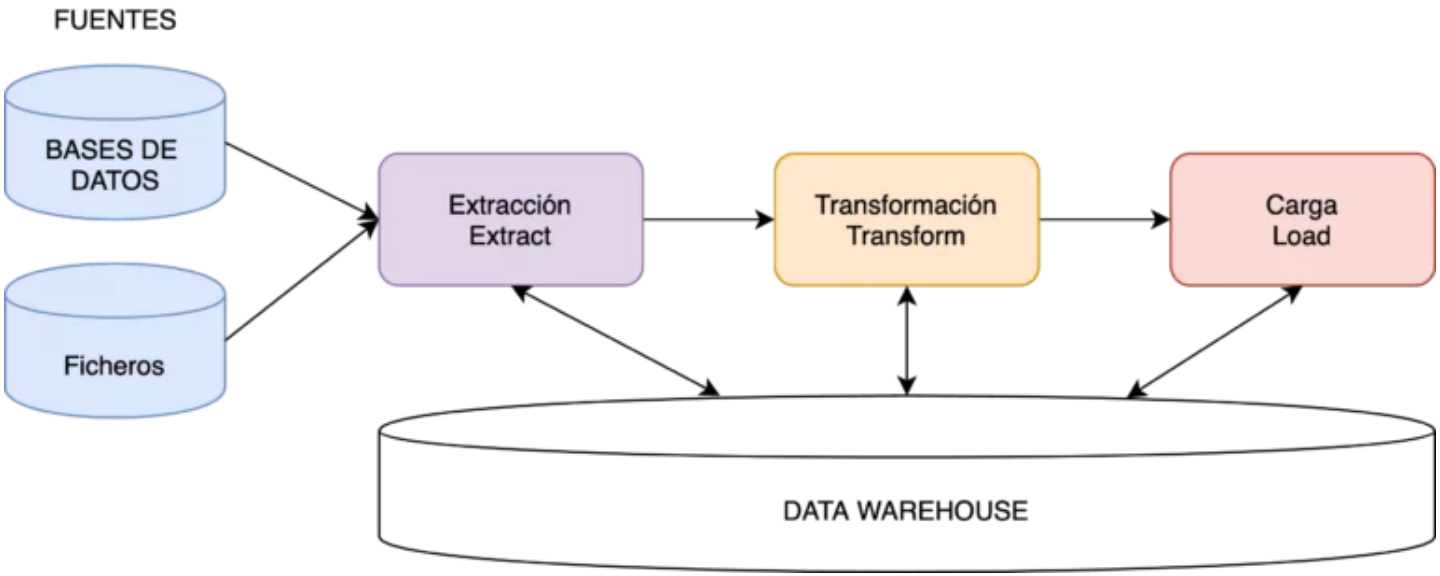
Another classic example is transforming our data into a format suitable for insertion into a database. Databases have their peculiarities, such as indexes or relationships, which may not be explicitly reflected in our data and often need to be generated through transformation and processing for the data insertion process into these external systems. Additionally, factors like character encoding and keywords need to be considered.

ETL-Based Mechanisms

ETL (Extract-Transform-Load) systems focus on mechanisms for interconnecting systems. In these systems, the key lies in the three phases, with a particular emphasis on the transformation phase, which is often the complex point. The description of the following points is as follows:

- *Extraction*: Retrieving data into machine-processable information.
- *Transformation*: Conversion, always focused on the future destination system, where the transformation process can enrich the data. The result may not necessarily be information but valid data in the destination format.
- *Load*: The process of converting data into a format that the destination system can interpret as information.

ETL-Based Mechanisms

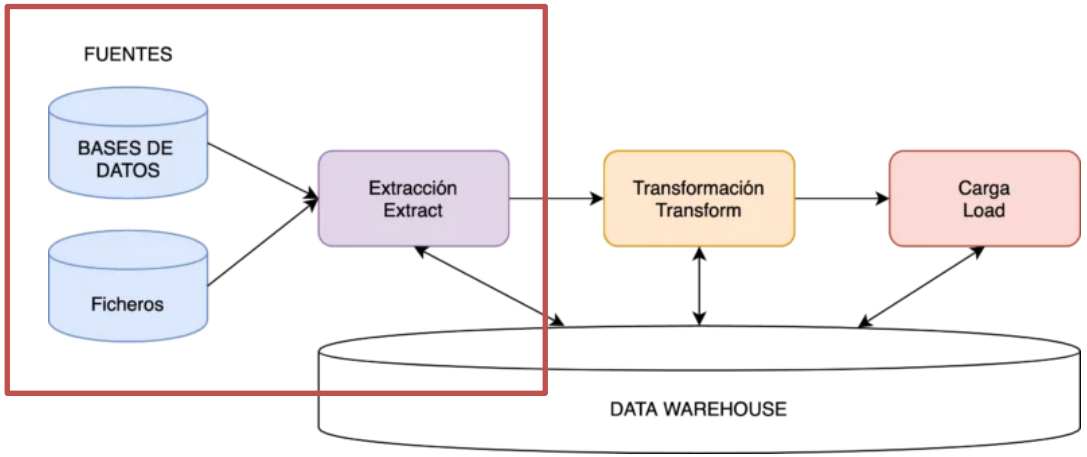


From: https://aprenderbigdata.com/

ETL-Based Mechanisms (Extract)

The extraction process is lighter than usual and allows for a significant reduction in the homogenization process, offloading all transformations to the post-processing transformation phase. This phase has the peculiarity that, at the end of it, there doesn't need to be consistent data, but primarily raw data in most cases.

This part of the process is often led by experts in the source of the data and its system.

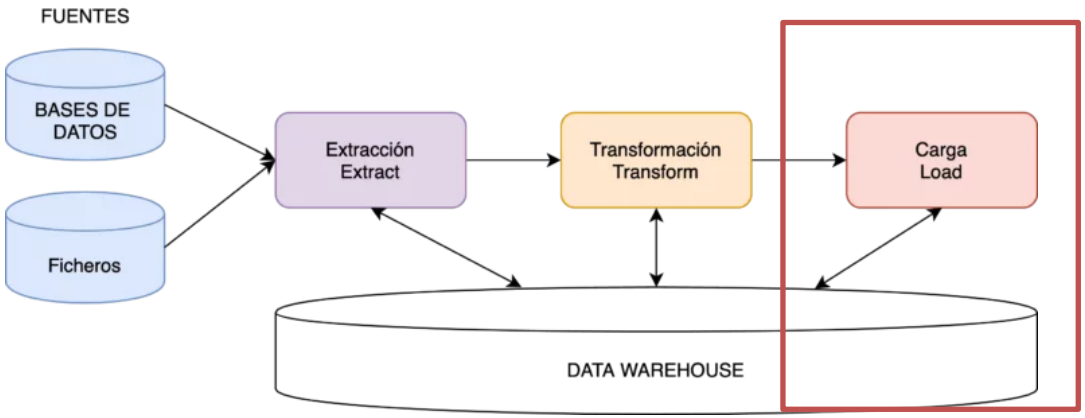


From: https://aprenderbigdata.com/

ETL-Based Mechanisms (Load)

The loading process is very similar to the extraction process. The objective here is to prioritize the data's quality, often abstracting information if present, to adapt it to the loading process in the new destination. This process focuses on ensuring that the data's interpretation by the destination format/program is knowledgeable about how to interpret it and create information.

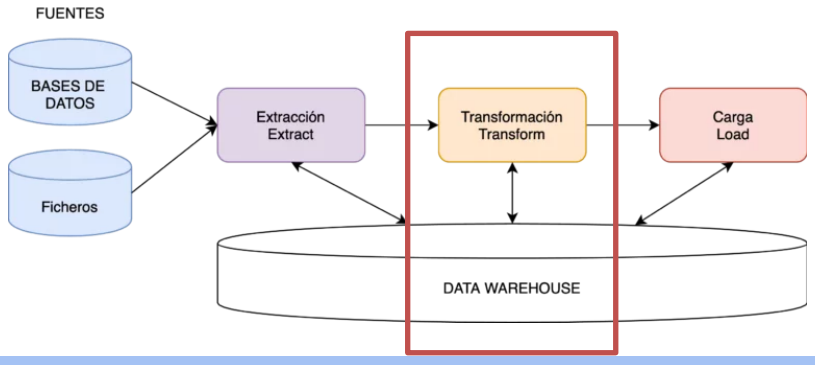
This part of the process is often led by experts in the destination of the data and its system.



From: https://aprenderbigdata.com/

ETL-Based Mechanisms (Transformation)

The transformation process plays a central role in these types of systems. While extraction and loading may only deal with data, the transformation process between the two systems considers multiple information-related concepts. The first is the need to interpret the extraction data into information that can be understood, processed, and potentially even expanded. This requires in-depth knowledge of the nature of the data and the underlying information, as well as the systems on both sides. Such transformations are typically led by highly qualified domain experts who can handle source systems, destination systems, data, and information, both implied and available.



From: https://aprenderbigdata.com/

ETL-Based Mechanisms (Advantages)

- In this type of data processing systems, the main burden falls on a single module, which is the transformation.
- While the workloads in extraction and loading are not high, they often involve labor-intensive processes that require a significant amount of time and a close understanding of the source/destination system.
- Although transformation is the most complex process, it can be relatively straightforward at the software level. There are well-studied techniques that allow the creation of robust software pipelines based on already implemented libraries, which simplifies the work in terms of the expert's time.
- The process is repeatable in multiple aspects. If you want to change the destination (loading) of the information, you only need to change part of the transformation and loading, not the entire system. Conversely, in the case of a new source, the solution is symmetrically similar. Additionally, many of the transformations are domain-dependent, so they tend to be reusable in most cases.
- Since transformations are more focused on the data model (semantics, syntax, dictionary), they can be processed in blocks.



ETL-Based Mechanisms (Disadvantages)

- Experts in the source and destination data are required for all three phases of the process.
- The transformation phase requires a deep understanding of the domain, which implies additional study of the domain by the expert.
- Solutions that cover 100% of current cases are feasible but rarely cover 100% of potential cases, so an error factor must be accepted in very complex cases.
- They may be theoretically viable but not computable in practice.
- They can have a variability factor that expands to two or more valid and equiprobable solutions. In such environments, the failure factor can be decisive.
- They do not guarantee that the pipeline will be reusable indefinitely. If one of the systems changes, the impact cannot be assessed until it is analyzed.



ETL-Based Mechanisms (Examples)

- Database transformation (e.g., MySQL to Oracle).
- Transformation between compatible formats (e.g., Word to OpenWriter).
- Updating files to previous or later versions (e.g., .doc to .docx).
- Migration between systems within the same domain (e.g., Drupal to WordPress).
- Compilers.
- Language (and programming language) translators.
- Applying filters to images.
- Noise reduction in waveforms.
- ...



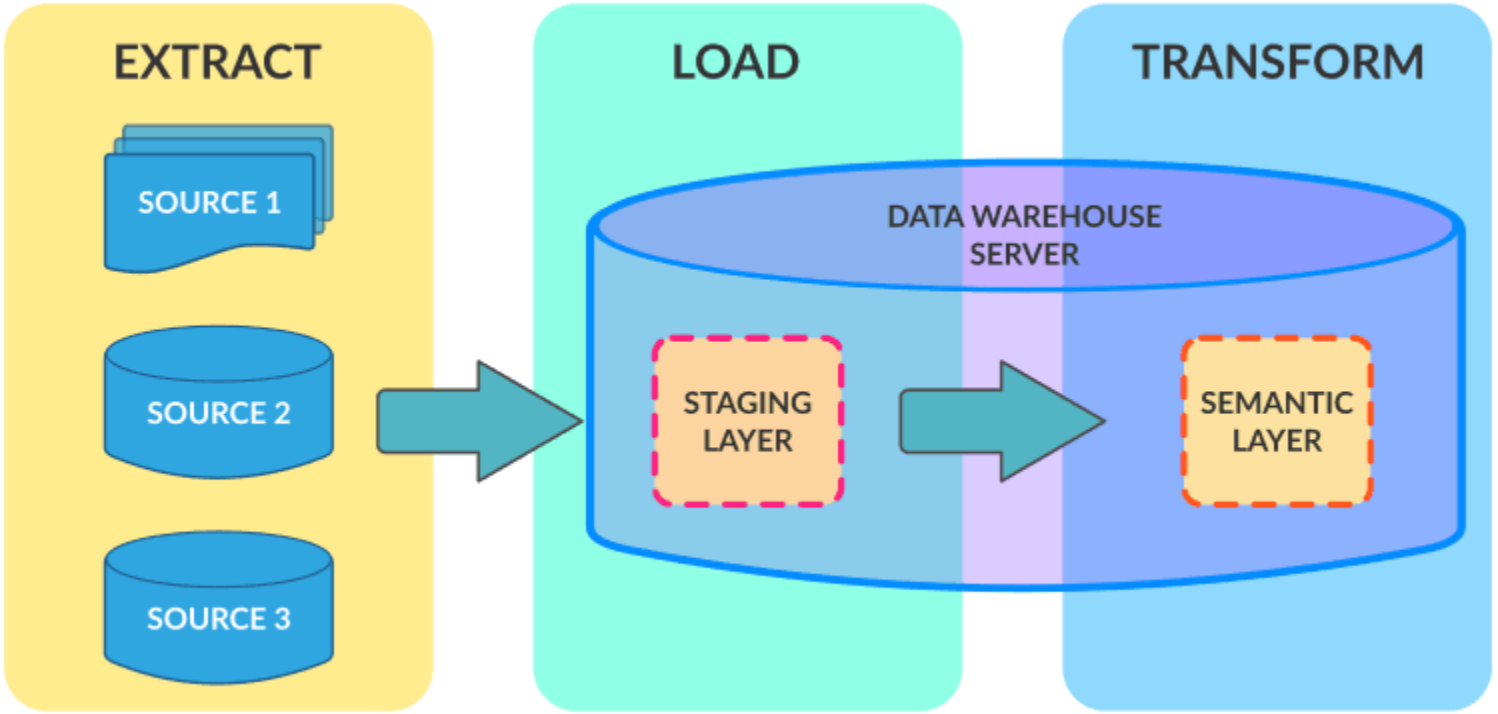
ELT-Based Mechanisms

The difference between ETL (Extract, Transform, Load) systems and ELT (Extract, Load, Transform) systems is that in ETL systems, the entire data set is processed to perform the transformation. In contrast, ELT systems focus on extracting data from data sources and loading it into a flexible storage system. Transformations are then applied directly to the data storage to make modifications and generate information.

In simple terms, the goal of ELT is not to create a final system but to make the storage system itself the final system.

As a result, ETL processes require managing the entire volume of data to extract information monolithically from them and re-store it in the system.

ELT-Based Mechanisms

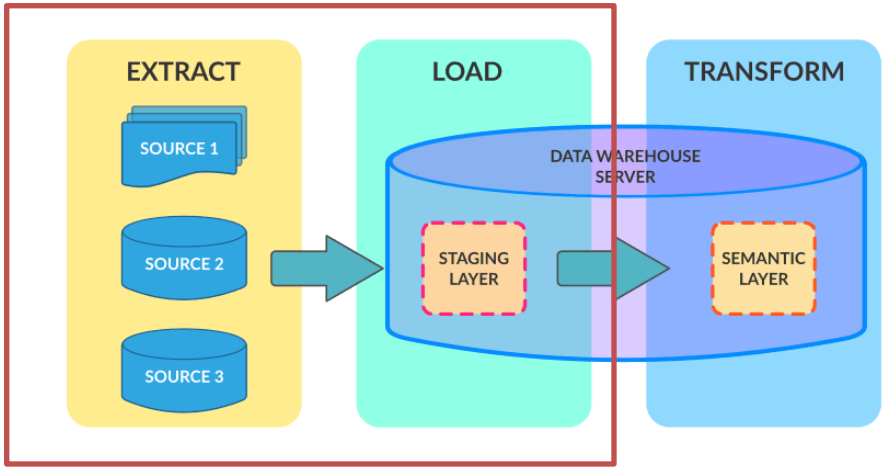


From: https://www.jet-software.com/

ELT-Based Mechanisms (phases)

- *Extract*: This phase is very similar to the phase in ETL processes, its objective once again is to extract data in a simple way without losing information, in many cases again, even in raw form.

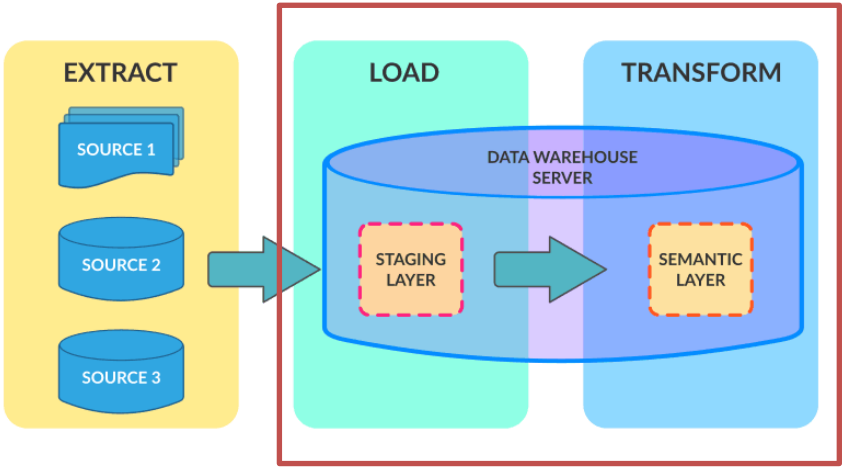
- *Load:* This phase, which in ETL is the final one, is focused on finding the simplest storage mechanism that allows us to extract data, transform it in a general way, and store it. In this type of loads, the option of storing in non-relational and document-oriented databases has become particularly strong in the market.



From: https://www.jet-software.com/

ELT-Based Mechanisms (phases)

- *Transform*: Again, the transformation phase becomes the most critical one, as its goal this time is to transform the collection of data/information into a new source of information using deep editing techniques. Once more, this phase requires a deep understanding of the data being processed and working meticulously on the transformation processes to ensure that no information is lost in any part of the transformation.



From: https://www.jet-software.com/

ELT-Based Mechanisms (Advantages)

- It can conduct deep processing by analyzing all data as a whole.
- It allows for deep changes to be made to a portion of the data as part of a more complex processing.
- It enables transformations that are more focused on "what" content is present rather than "how" it's presented.
- It facilitates cyclic transformation processes, with or without learning, which continue until there's no new transformation to apply.
- It allows for future transformations to be applied.



ELT-Based Mechanisms (Disadvantages)

- In case of an error, tracing the point of failure is more challenging.
- It requires a significant amount of storage and substantial read/write operations.
- It still faces the same issues as ETL, where the domain expert must have a thorough knowledge of the sources.
- It extends the challenge for the expert to understand the data and the different structures that are temporarily generated in the system during the transformation process.
- High-performance computers are required to perform complex calculations on extensive datasets.
- The difficulty of transformation is directly proportional to the size of the data and the analysis required for transformation, and it becomes exponentially challenging with larger datasets.



ELT-Based Mechanisms (Examples)

- Information extraction from texts.
- Information analysis and data inference.
- Search for patterns and final states within datasets.
- Deep Learning.
- Neural Networks.
- …



Classic Transformations

- Translation, Mapping, Parsing: An element or value should correspond to a sequence or regular expression and is replaced by another equivalent value or expression.
  - Uses cases: Schema translation, value translation, equivalence between simple structures, synonym removal/sustitution, etc.

Classic Transformations

- Union and Separation: Based on the structure or values, it is processed for the union of these or the separation into various different values. These types of processes occur in structures with multi-valued information across several fields or within a single field.
  - Use cases: Unifying several repetitions of a value to generate a unique continuous structure for the value (such as arrays) or multiple pieces of information, such as complete addresses separated by address fields, floor, postal code, etc.

Classic Transformations

- Cleaning, Filtering, and Sorting: Information Reduction Mechanisms for eliminating low-value information, indexing, and sorting of the most relevant data at the beginning of documents to promote, present, the most in-depth information in the highlighted area. Reordering values or categories to create sections based on order or highlight values within the information.
  - Use cases: Use cases: Sorting categories, removal of indexes, creation of search indexes, document titling, visual document reading improvement, etc.

Advanced/Deep Transformations

- Model-driven transformation: Model-driven transformations are used when the structure of the data is governed by a model. These transformations analyze the data structure and make changes, recombine elements, or create new ones based on this structure. They typically employ formal grammar trees to process the structure and carry out the transformation.
  - Use cases: Model-driven transformations are commonly used in compilation processes, interpreted code execution (like PHP), or structured file transformations (such as XMLT for XML), etc.

Advanced/Deep Transformations

- Data-driven transformations: focus on processing data based on the values themselves. These transformations don't rely on a predefined structure but instead analyze and manipulate data based on its content and specific patterns. They are highly adaptable and flexible, making them suitable for handling data that doesn't conform to a strict model.
  - Use cases: Data-driven transformations are often used in natural language processing, pattern recognition, and data analysis, among other applications..



POLITÉCNICA

UNIVERSIDAD
POLITÉCNICA
DE MADRID



Universidad
Politécnica
de Madrid

ETSI **SISTEMAS INFORMÁTICOS**